iND83213A CS01 Bootloader Update User Guide V1.04

Preliminary

1 REVISION HISTORY

Table 1 Revision History

Rev #	Date	Action	By
1.00	10/9/2023	First draft Rev: libdev_iND83213A_CS01_Bootloader.B0.01.03	Winter
1.01	2023/11/2	Rev: libdev_iND83213A_CS01_Bootloader.B0.01.04 (1) SCC Disable (2) make sure lin_swon = '1' and fast slew rate (3) code clean	Winter
1.02	1/29/2024	Rev: libdev_iND83213A_CS01_Bootloader.B0.01.05 (1)Add a new function for requiring 3D response when no 3C request	EricChen
1.03	3/21/2024	Rev: libdev_iND83213A_CS01_Bootloader.B0.01.06 (1) Update LIN Performance Params for LIN_AA	Winter
1.04	4/2/2024	Rev: libdev_iND83213A_CS01_Bootloader.B0.01.07 (1) Update LIN Performance Params for LIN_AA (2) Fix PWM device API error (default status these API no be called)	Winter

Preliminary

2 TABLE OF CONTENTS

1 REVISION HISTORY 2

2 TABLE OF CONTENTS 3

3 V1.04 4

3.1 Version Change 4

3.2 BUG FIXED 4

3.2.1 Update LIN device Params for improve LIN_AA performance 4

3.2.2 Fix PWM device API EROR 5

4 V1.03 6

4.1 Version Change 6

4.2 BUG FIXED 6

4.2.1 Update LIN device Params for improve LIN_AA performance 6

5 V1.02 7

5.1 Version Change 7

5.2 New Feature 7

5.2.1 Require 3D response when no 3C request 7

6 V1.01 8

6.1 Version Change 8

6.1.1 Disable SCC in Bootloader Project 8

6.1.2 Make Sure SWON=1 and config fast LIN slew rate in Bootloader Project 8

6.2 Clean CODE 9

7 V1.00 10

7.1 Version Change 10

7.2 BUG Fixed 10

7.2.1 After Chip reset, Firmware load LDO trim code to register immediately 10

7.3 New Features 10

Preliminary

3 V1.04

3.1 VERSION CHANGE

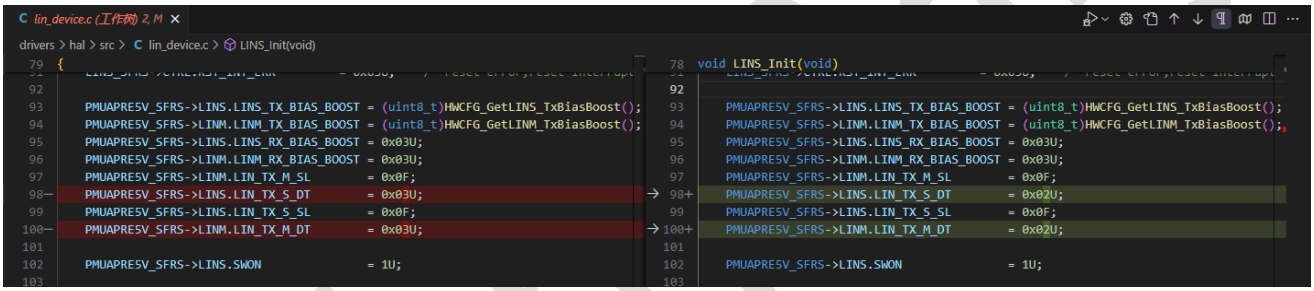
Form Version: libdev_iND83213A_CS01_Bootloader.B0.01.06

Update Version: libdev_iND83213A_CS01_Bootloader.B0.01.07

3.2 BUG FIXED

3.2.1 Update LIN device Params for improve LIN_AA performance

New configuration can improve LIN Duty after multiple nodes.



TestData: 220pF @LININ and 220pF @LINOUT VBAT=13.5V. 16 EVK. condition.

Num of 213ACS01EVK = 16;VBAT = 13V5					
Conditon	FID	SamplePoint：Node1 slave data field		SamplePoint：Nodex master sync field	
		duty(us)	duty(%)	duty(us)	duty(%)
VBAT=13.5V LINS_RX BIAS_BOOST =3 LINM_RX_BIAS_BOOST = 3 LINS_SLEWRATE = 0xF LINM_SLEWRATE = 0xF LINS_TX_S_DT = 2 LINM_TX_S_DT = 2	0x10	52.41	50.39	50.92	48.96
	0xF	52.82	50.79	50.97	49.01
	0xE	52.4	50.38	50.89	48.93
	0xD	52.4	50.38	50.81	48.86
	0xC	52.79	50.76	50.97	49.01
	0xB	53.24	51.19	51.01	49.05
	0xA	52.92	50.88	50.92	48.96
	0x9	53.25	51.20	50.88	48.92
	0x8	53.42	51.37	50.71	48.76
	0x7	52.4	50.38	50.92	48.96
	0x6	53.19	51.14	50.76	48.81
	0x5	53.19	51.14	50.81	48.86
	0x4	53.19	51.14	51.18	49.21
	0x3	52.98	50.94	50.92	48.96
	0x2	52.5	50.48	51.03	49.07
	0x1	52.5	50.48	51.03	49.07

3.2.2 Fix PWM device API EROR

pwm_device.c (工作树) 2, M

drivers > hal > src > pwm_device.c > PWM_Handler(void)

```c
static uint16_t currPWM[24U];
static PwmIsrCallback_t pwmReloadCallback = NULL;

void PWM_Handler(void)
{
    if (pwmReloadCallback == NULL){
        pwmReloadCallback();
    }
}
```

```c
static uint16_t currPWM[24U];
static PwmIsrCallback_t pwmReloadCallback = NULL;

void PWM_Handler(void)
{
    if (pwmReloadCallback != NULL){
        pwmReloadCallback();
    }
}
```

pwm_device.c (工作树) 2, M

drivers > hal > src > pwm_device.c > PWM_Handler(void)

```c
/**
 */
void PWM_Init(PwmPrescaler_t divide, uint16_t period)
{
    IOCTRLA_SFRS->HWMODE = 0x0FFFU;
    PWM_SFRS->BASE.PRESCALESEL = (uint8_t)divide;
    PWM_SFRS->BASE.PERIOD = period;

    /* set duty cycle to 0*/
    for (uint8_t i = 0U; i < 24U; i++){
        PWM_SFRS->PULSE[i].PFALL = PWM_VALUE_MAX;
        PWM_SFRS->PULSE[i].PRISE = PWM_VALUE_MAX;
    }
    PWM_StartAndUpdate();
}
```

```c
 * @brief The instance of PMW device
 */
void PWM_Init(PwmPrescaler_t divide, uint16_t period)
{
    IOCTRLA_SFRS->HWMODE = 0xFFFFFFU;
    PWM_SFRS->BASE.PRESCALESEL = (uint8_t)divide;
    PWM_SFRS->BASE.PERIOD = period;

    /* set duty cycle to 0*/
    for (uint8_t i = 0U; i < 24U; i++){
        PWM_SFRS->PULSE[i].PFALL = PWM_VALUE_MAX;
        PWM_SFRS->PULSE[i].PRISE = PWM_VALUE_MAX;
    }
    PWM_StartAndUpdate();
}
```

pwm_device.c (工作树) 2, M

drivers > hal > src > pwm_device.c > PWM_Handler(void)

```c
/**
 */
void PWM_StartAndUpdate(void)
{
    PWM_SFRS->ENAREQ.ENAREQALL = 1U;

    PWM_SFRS->UPDATE = 1U;
}
```

```c
 * @brief The instance of PMW device
 */
void PWM_StartAndUpdate(void)
{
    PWM_SFRS->ENAREQ.ENAREQALL = 1U;
    PWM_SFRS->ENAREQ.ENAREQ = 0xFFFFFF;
    PWM_SFRS->UPDATE = 1U;
}
```

pwm_device.c (工作树) 2, M

drivers > hal > src > pwm_device.c > PWM_Handler(void)

```c
/**
 */
void PWM_RegisterPWMReloadIRQ(PwmIsrCallback_t callback)
{
    pwmReloadCallback = callback;
    PWM_SFRS->INTPWM.CLEAR.PERIOD = 1U;
    PWM_SFRS->INTPWM.IRQ.PERIOD = 1U;
    PWM_SFRS->UPDATE = 1U;
    NVIC_EnableIRQ(PWM_IRQn);
}

/**
 * @brief The instance of PMW device
 */
void PWM_UnRegisterPWMReloadIRQ(void)
{
    pwmReloadCallback = NULL;
    PWM_SFRS->INTPWM.CLEAR.PERIOD = 1U;
    PWM_SFRS->INTPWM.IRQ.PERIOD   = 0U;
    PWM_SFRS->UPDATE = 1U;
    NVIC_DisableIRQ(PWM_IRQn);
}
```

```c
 * @brief The instance of PMW device
 */
void PWM_RegisterPWMReloadIRQ(PwmIsrCallback_t callback)
{
    pwmReloadCallback = callback;
    PWM_SFRS->INTPWM.CLEAR.PERIOD = 1U;
    PWM_SFRS->INTPWM.ENABLE.PERIOD = 1U;
    PWM_SFRS->UPDATE = 1U;
    NVIC_EnableIRQ(PWM_IRQn);
}

/**
 * @brief The instance of PMW device
 */
void PWM_UnRegisterPWMReloadIRQ(void)
{
    pwmReloadCallback = NULL;
    PWM_SFRS->INTPWM.CLEAR.PERIOD = 1U;
    PWM_SFRS->INTPWM.ENABLE.PERIOD   = 0U;
    PWM_SFRS->UPDATE = 1U;
    NVIC_DisableIRQ(PWM_IRQn);
}
```

4 V1.03

4.1 Version Change

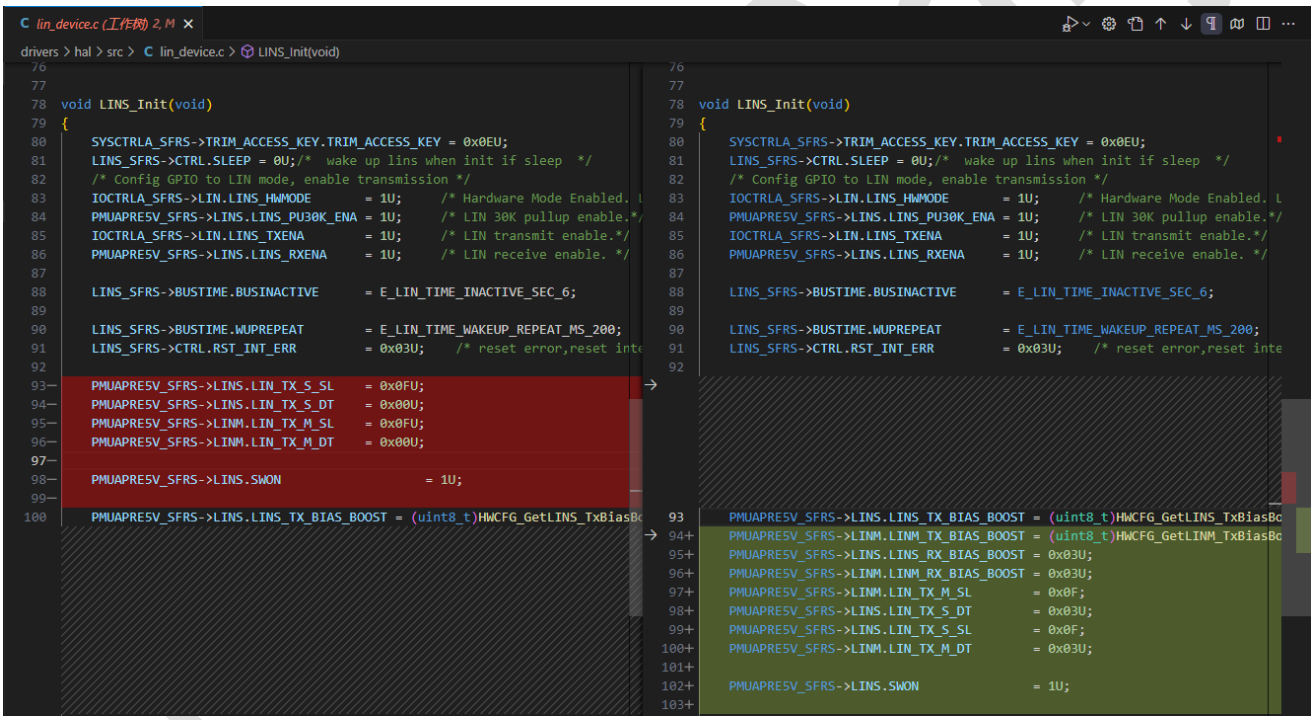
Form Version: libdev_iND83213A_CS01_Bootloader.B0.01.05

Update Version: libdev_iND83213A_CS01_Bootloader.B0.01.06

4.2 BUG FIXED

4.2.1 Update LIN device Params for improve LIN_AA performance

New configuration can improve LIN Duty after multiple nodes.



5 V1.02

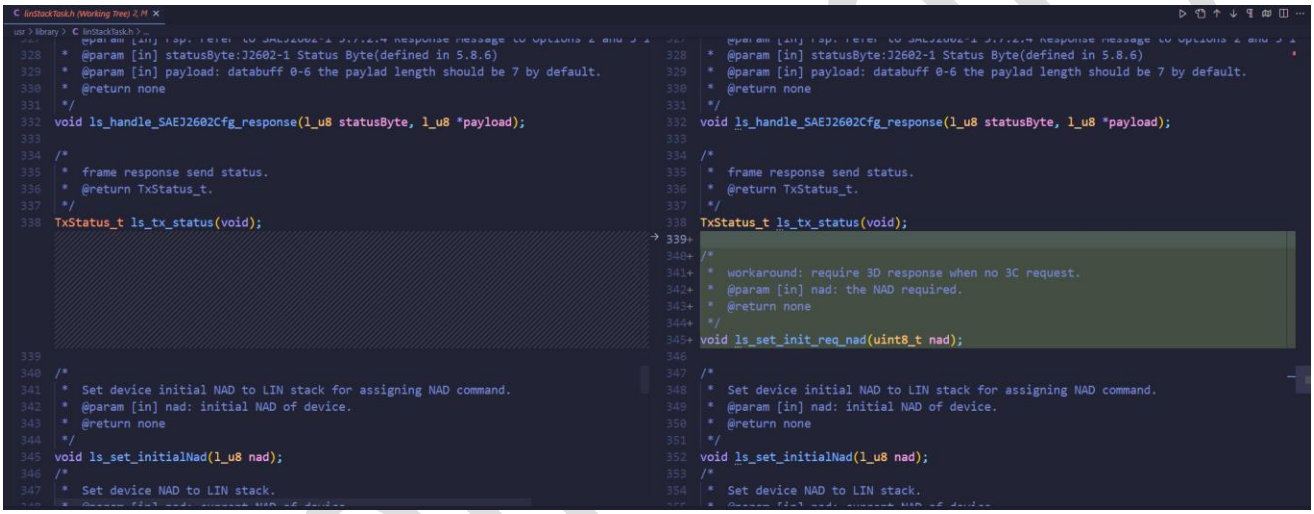
5.1 Version Change

Form Version: libdev_iND83213A_CS01_Bootloader.B0.01.04

Update Version: libdev_iND83213A_CS01_Bootloader.B0.01.05

5.2 New Feature

5.2.1 Require 3D response when no 3C request



Preliminary

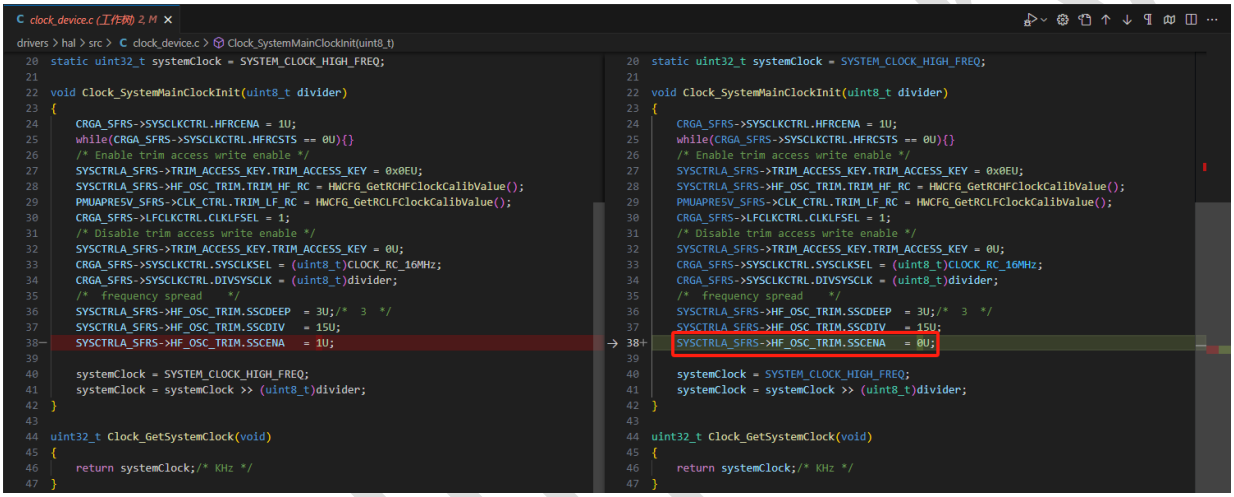
6 V1.01

6.1 Version Change

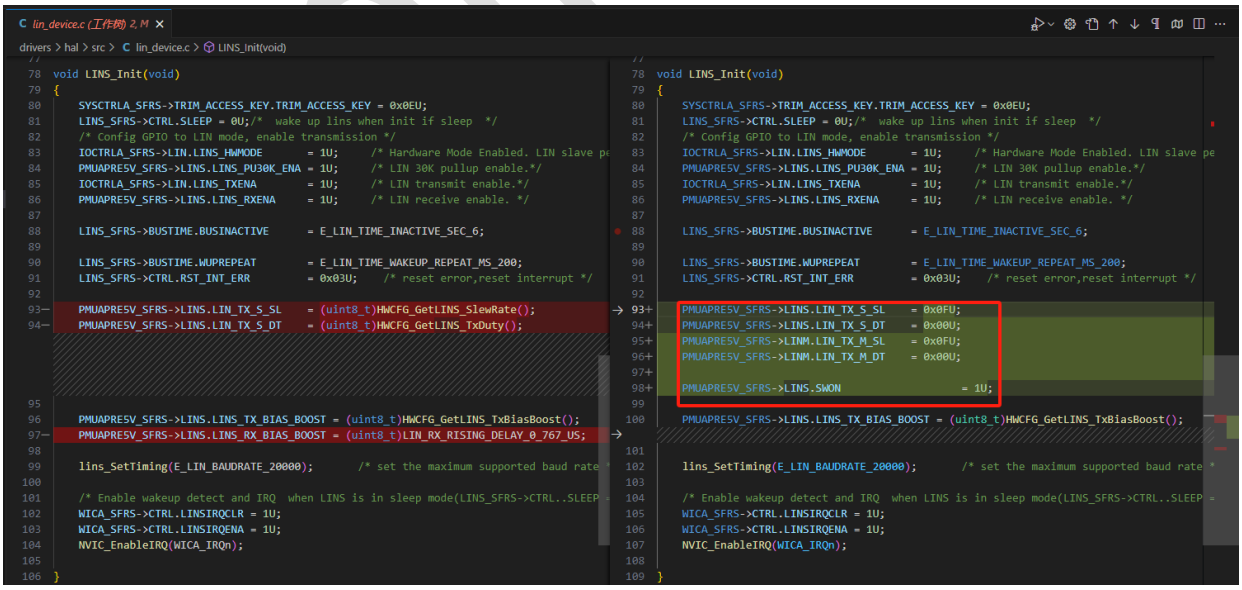
Form Version: libdev_iND83213A_CS01_Bootloader.B0.01.03

Update Version: libdev_iND83213A_CS01_Bootloader.B0.01.04

6.1.1 Disable SCC in Bootloader Project

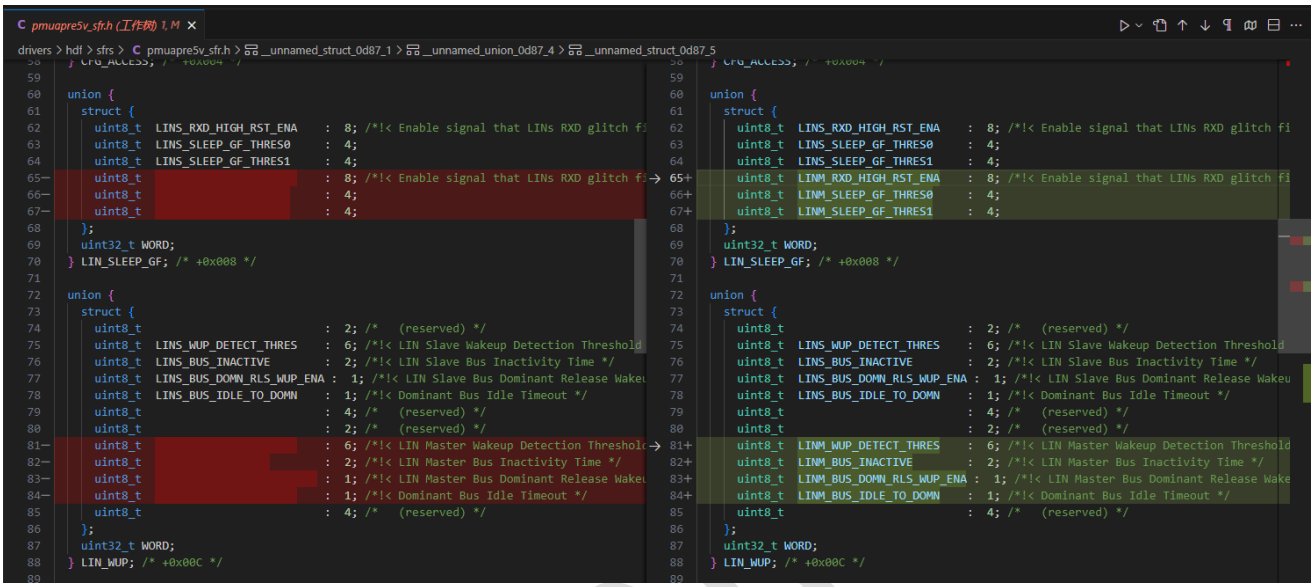


6.1.2 Make Sure SWON=1 and config fast LIN slew rate in Bootloader Project



6.2 CLEAN CODE

Modify head files code from app project. For Example:



Prelim

7 V1.00

7.1 Version Change

Form Version: libdev_iND83213A_CS01_Bootloader.B0.01.02

Update Version: libdev_iND83213A_CS01_Bootloader.B0.01.03

7.2 BUG Fixed

7.2.1 After Chip reset, Firmware load LDO trim code to register immediately

```
main.c (工作区) 2, M
usr > main.c > main(void)

void main(void)
{
    /* if it's wakeup signal, program should jump to application directly */
    if ( PUM_IsWakupSignal()== TRUE){
        APP_JumpToApplication();
    }
    /* !!!!!!!MUST BE called firstly here for initializing system parameters !!!!*/
    PDS_Init();
    /* System init for hardwre init */
    SYS_Init();
    /* system main infinite loop */
    for(;;){
      TM_RunTasks();
    }
}
```

```
void main(void)
{
    /* Load TrimCode to ASIC */
    PMU_Init();
    /* if it's wakeup signal, program should jump to application directly */
    if ( PUM_IsWakupSignal()== TRUE){
        APP_JumpToApplication();
    }
    /* !!!!!!!MUST BE called firstly here for initializing system parameters !!!!*/
    PDS_Init();
    /* System init for hardwre init */
    SYS_Init();
    /* system main infinite loop */
    for(;;){
      TM_RunTasks();
    }
}
```

```
pmu_device.c (工作区) 2, M
drivers > hal > src > pmu_device.c > PMU_Init(void)

}
```

```
}
void PMU_Init(void)
{
    SYSCTRLA_SFRS->PMU_ACCESS_KEY.PMU_ACCESS_KEY        = 0x0AU;
    SYSCTRLA_SFRS->TRIM_ACCESS_KEY.TRIM_ACCESS_KEY      = 0x0EU;
    PMUAPRE5V_SFRS->CFG_ACCESS.CFG_ACCESS_KEY           = 0x0BU;

    PMUAPRE5V_SFRS->CTRL.TRIM_BG_V2I     = HWCFG_Get1V2TrimCode();
    PMUA_SFRS->PMUTRIM.VDD1V5_LDO_TRIM   = HWCFG_GetLDO1V5TrimCode();
    PMUA_SFRS->PMUTRIM.VDD3V3_LDO_TRIM   = HWCFG_GetLDO3V3TrimCode();
    PMUA_SFRS->PMUTRIM.OCP_CTRL_1V5      = HWCFG_GetLDO1V5OCPTrimCode();
    PMUA_SFRS->PMUTRIM.OCP_CTRL_3V3      = HWCFG_GetLDO3V3OCPTrimCode();
    PMUA_SFRS->PMUTRIM.OCP_CTRL_5V0      = HWCFG_GetLDO5V0OCPTrimCode();
}
```

7.3 New Features